

Unit 1

Design Thinking as Mindset

3 Slibraries · 15 framework sheets

SLIBRARY 1 · DT Foundations

5 sheets

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| 1 Brown's Triple Loop | 4 DVF Framework |
| 2 Stanford 5-Stage | 5 3-i Framework |
| 3 Double Diamond | |

SLIBRARY 2 · DT Mindsets

5 sheets

- | | |
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| 1 Liedtka's 4 Truths | 4 Buchanan's Wicked Problems |
| 2 Dosi's 10-Dim Mindset | 5 Mindset · Process · Toolbox |
| 3 Kelley's Creative Confidence | |

SLIBRARY 3 · DT Process & Tools

5 sheets

- | | |
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| 1 Empathy Map | 4 5 Whys |
| 2 Personas | 5 How Might We (HMW) |
| 3 Customer Journey Map | |

Brown's Triple Loop

SLIBRARY 1 · SHEET 1 / 5

Inspiration → Ideation → Implementation. The three spaces every DT project cycles through

PROFESSIONAL DEFINITION

Tim Brown's foundational account of design thinking organises the work into three overlapping spaces — Inspiration, Ideation and Implementation — rather than a linear sequence of steps. Inspiration surfaces the problem through context and empathy; Ideation generates and prototypes solutions; Implementation brings the chosen solution to life. Projects loop back through the spaces as learning accumulates, and any space can be re-entered when new evidence arrives (Brown, 2008; Brown and Wyatt, 2010).

WHO DEVELOPED IT

Articulated by Tim Brown, then CEO of IDEO, in his 2008 *Harvard Business Review* article *Design Thinking*, drawing on three decades of IDEO consulting practice and David Kelley's earlier Stanford and IDEO methodology.

WHY IT WAS DEVELOPED

Brown wanted to liberate design from a phase-gate process metaphor — which mis-framed it as a final "styling" step — and re-position it as a mindset for innovation. Treating the work as overlapping spaces makes iteration a feature, not a bug, of doing it well.

HOW IT IS USED

Teams move fluidly between the three spaces, with frequent looping. Each space uses different tools (ethnographic field-work, brainstorming, prototyping) but shares one rule: everything is provisional until evidence arrives.

WHEN IT IS USED

Use early in any innovation project where the problem is not yet well-defined, when re-framing matters more than execution speed, and whenever a team is tempted to skip from Inspiration straight to Implementation.

BY WHOM IT IS USED

IDEO consultants, in-house innovation teams, MBA design-thinking courses, and senior managers leading multi-disciplinary teams in product, service and policy contexts.

FIGURE 1 · BROWN'S THREE SPACES

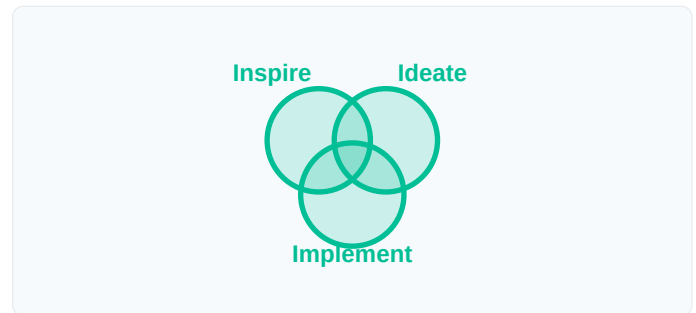


Figure 1. The three spaces overlap; projects loop back as learning accumulates. Adapted from Brown (2008).

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Where Herbert Simon (1969) framed design as iterative problem-solving and the Stanford d.school's 5-stage process (c. 2005) named distinct phases, Brown's contribution was to frame the work as *overlapping* rather than *stepwise*. The triple-space mental model gave non-designers permission to revisit earlier work — the move that opened design thinking to general management.

RELATED FRAMEWORKS IN THIS COURSE

DVF Framework · Slibrary 1

Stanford 5-Stage · Slibrary 1

Double Diamond · Slibrary 1

3-i Framework · Slibrary 1

SOURCES (HARVARD REFERENCING STYLE)

Brown, T. (2008) 'Design Thinking', *Harvard Business Review*, 86(6), pp. 84–92.

Brown, T. and Wyatt, J. (2010) 'Design Thinking for Social Innovation', *Stanford Social Innovation Review*, 8(1), pp. 30–35.

Brown, T. (2009) *Change by Design*. New York: HarperBusiness.

Kelley, T. and Littman, J. (2001) *The Art of Innovation*. New York: Doubleday.

Stanford 5-Stage

SLIBRARY 1 · SHEET 2 / 5

Empathize · Define · Ideate · Prototype · Test. The d.school's canonical process

PROFESSIONAL DEFINITION

The d.school's canonical teaching framework — Empathize, Define, Ideate, Prototype, Test — is presented as a linear diagram but is explicitly non-linear in practice. Teams revisit earlier stages as testing reveals new insight, using the stages as *modes of thinking* rather than project phases. The framework's pedagogical clarity (five named modes) made it the most-taught design-thinking model worldwide (Plattner, 2010; Plattner, Meinel and Leifer, 2011).

WHO DEVELOPED IT

Codified at the Hasso Plattner Institute of Design at Stanford (the d.school) by Hasso Plattner, David Kelley and colleagues from 2005 onwards. Documented in the d.school's *Bootcamp Bootleg* (2010) and *An Introduction to Design Thinking: Process Guide* (Plattner, 2010).

WHY IT WAS DEVELOPED

Brown's (2008) overlapping spaces were powerful conceptually but hard to teach. The d.school broke the work into five named modes that students could practise, recognise and switch between deliberately — making design thinking transmissible at scale across MBA programmes, executive education and K-12.

HOW IT IS USED

Walk the five modes in order on first pass, then loop back as needed. Each mode has its own deliverables: empathy maps, point-of-view statements, ideation lists, prototypes, test plans. Modes are switched deliberately, not unconsciously.

WHEN IT IS USED

Use as a teaching scaffold for newcomers to design thinking, in workshop facilitation where named handoffs help, and in regulated environments where stage gates are required by governance.

BY WHOM IT IS USED

Design educators, workshop facilitators, MBA students, and corporate-innovation programme directors using a structured curriculum. Standard scaffold in over 1,000 universities worldwide.

FIGURE 1 · FIVE MODES

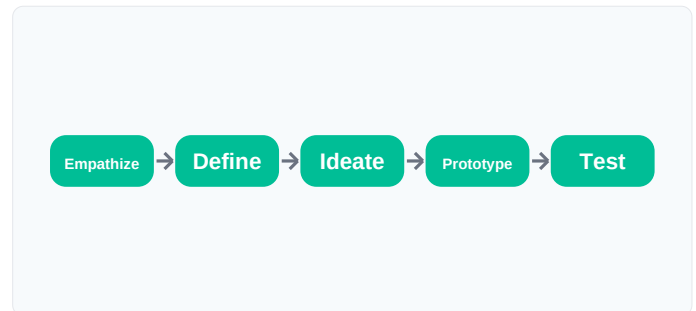


Figure 1. Five modes are walked in order on first pass, then re-entered as evidence demands.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Where Brown (2008) framed design thinking as overlapping spaces and IDEO HCD (2009) used a three-lens triangle, the d.school's contribution is the named-mode pedagogy that makes the work teachable. Building on Simon's (1969) iterative-problem-solving tradition, the 5-stage process became the canonical entry-level model.

RELATED FRAMEWORKS IN THIS COURSE

[Brown's Triple Loop · Slibrary 1](#)
[Double Diamond · Slibrary 1](#)
[DVF Framework · Slibrary 1](#)
[3-i Framework · Slibrary 1](#)

SOURCES (HARVARD REFERENCING STYLE)

Plattner, H. (2010) *An Introduction to Design Thinking: Process Guide*. Stanford: Hasso Plattner Institute of Design at Stanford.
 Plattner, H., Meinel, C. and Leifer, L. (eds.) (2011) *Design Thinking: Understand – Improve – Apply*. Berlin: Springer.
 d.school (2010) *Bootcamp Bootleg*. Stanford: Hasso Plattner Institute of Design.
 Brown, T. (2009) *Change by Design*. New York: HarperBusiness.

Double Diamond

SLIBRARY 1 · SHEET 3 / 5

Discover · Define · Develop · Deliver. Two diamonds of divergence and convergence

PROFESSIONAL DEFINITION

The UK Design Council's Double Diamond is a 2005 mapping of the creative process as two diamonds of divergence and convergence: *Discover* (explore the problem widely), *Define* (frame it sharply), *Develop* (explore solutions widely), *Deliver* (test and launch). The first diamond — by forcing problem-divergence before convergence — prevents teams from solving the wrong problem, the most common and expensive failure mode in design (UK Design Council, 2005, 2019).

WHO DEVELOPED IT

The UK Design Council, building on Béla Bánáthy's (1996) divergence-convergence diagrams and the broader systems-design tradition. Refreshed in 2019 as the "Framework for Innovation" with explicit feedback loops between the four phases.

WHY IT WAS DEVELOPED

Design Council researchers studying eleven major design firms found that the highest-performing teams alternated divergent and convergent thinking deliberately, with explicit hand-offs at each transition. The Double Diamond externalised this pattern as a teachable visual — the most-shared design-process diagram in the world.

HOW IT IS USED

Open the first diamond wide (Discover): research, observe, gather. Close it sharply (Define): one specific problem statement. Open the second diamond (Develop): generate and prototype options. Close it (Deliver): test, refine, launch.

WHEN IT IS USED

Use whenever the problem is not yet well-defined, in service-design and policy-design contexts, and as a senior-leadership communication tool for explaining design's value over straight-to-solution thinking.

BY WHOM IT IS USED

Service designers, policy designers, public-sector innovation teams, agency design leaders, and senior managers explaining design investment to executive sponsors.

FIGURE 1 · TWO DIAMONDS

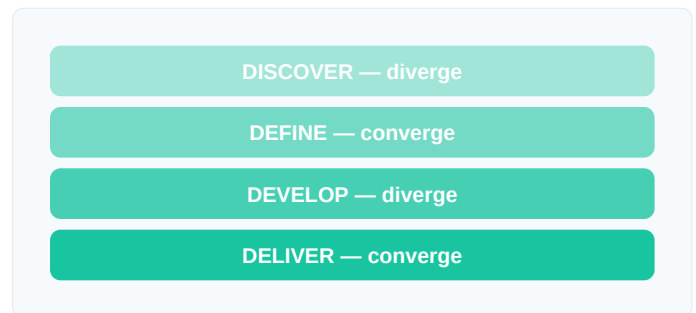


Figure 1. The first diamond prevents solving the wrong problem; the second prevents converging on a weak solution.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Where Brown's (2008) Triple Loop named overlapping spaces and the Stanford 5-stage named modes, the Double Diamond's contribution is making the divergent-convergent rhythm visible. Building on Bánáthy's (1996) systems-design diagrams, it gave teams a shared map for explicit thinking-mode hand-offs.

RELATED FRAMEWORKS IN THIS COURSE

[Brown's Triple Loop · Slibrary 1](#)
[Stanford 5-Stage · Slibrary 1](#)
[DVF Framework · Slibrary 1](#)
[3-i Framework · Slibrary 1](#)

SOURCES (HARVARD REFERENCING STYLE)

UK Design Council (2005) *A Study of the Design Process: Eleven Lessons*. London: Design Council.

UK Design Council (2019) *Framework for Innovation*. London: Design Council.

Bánáthy, B.H. (1996) *Designing Social Systems in a Changing World*. New York: Plenum.

Brown, T. (2009) *Change by Design*. New York: HarperBusiness.

DVF Framework

SLIBRARY 1 · SHEET 4 / 5

Desirability × Feasibility × Viability — the innovation sweet spot

PROFESSIONAL DEFINITION

IDEO's DVF Framework is the innovation sweet-spot heuristic: an idea is worth pursuing only where *Desirability* (people want it), *Feasibility* (we can build it), and *Viability* (the business sustains it) intersect. The framework organises every design-thinking decision as a test of evidence across these three dimensions. Ideas that touch only two of the three lenses are flagged as incomplete: needing feasibility work, market evidence, or a viable business model (Brown, 2009; IDEO.org, 2015).

WHO DEVELOPED IT

Tim Brown and the IDEO team, codified in *Change by Design* (Brown, 2009) and operationalised through IDEO's social-sector arm IDEO.org (2015). The three-circle framing has roots in Larry Keeley's (Doblin) earlier innovation work and in IDEO's 1990s consulting practice.

WHY IT WAS DEVELOPED

Innovation teams over-optimised on whichever lens fit their disciplinary background — engineers chased feasibility, marketers chased desirability, finance chased viability — and shipped incomplete products. DVF made all three dimensions explicit and equal, forcing cross-functional debate about which lens needed more evidence.

HOW IT IS USED

For every candidate idea, audit the evidence behind each lens. Score qualitatively (strong / moderate / weak / unknown). Only ideas with strong evidence on all three lenses pass the gate; ideas weak on one lens trigger targeted research before commitment.

WHEN IT IS USED

Use in stage-gate reviews, in venture-capital and corporate-innovation portfolio screens, in strategy-of-innovation workshops, and when teams default to single-lens thinking.

BY WHOM IT IS USED

Innovation portfolio managers, design and product leadership, venture investors, corporate-innovation programme directors, and consultants advising on innovation governance.

FIGURE 1 · THREE-LENS SWEET SPOT

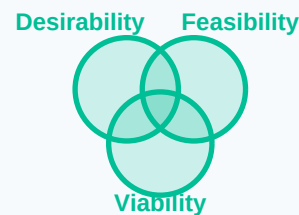


Figure 1. Innovation lives at the intersection; ideas weak on any one lens need targeted research, not commitment.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Where Cooper's (1990) Stage-Gate process used many criteria with weights and Pugh's (1991) concept selection used multi-criteria matrices, DVF's contribution is the radical simplification to three dimensions. Building on Doblin's innovation typology (Keeley et al., 2013), it became the canonical innovation lens of the design-thinking era.

RELATED FRAMEWORKS IN THIS COURSE

Brown's Triple Loop · Slibrary 1

3-i Framework · Slibrary 1

Stanford 5-Stage · Slibrary 1

Double Diamond · Slibrary 1

SOURCES (HARVARD REFERENCING STYLE)

- Brown, T. (2009) *Change by Design*. New York: HarperBusiness.
 IDEO.org (2015) *The Field Guide to Human-Centered Design*. San Francisco: IDEO.org.
 Keeley, L., Walters, H., Pikkell, R. and Quinn, B. (2013) *Ten Types of Innovation*. Hoboken: Wiley.
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3-i Framework

SLIBRARY 1 · SHEET 5 / 5

Insight · Idea · Impact. IDEO's compressed three-step account of design work

PROFESSIONAL DEFINITION

The 3-i Framework is IDEO's condensed three-step account — *Insight, Idea, Impact* — used when a full 5-stage map is too heavy. *Insight* is the observed human truth uncovered through field research; *Idea* is the response to it (concept, prototype or service); *Impact* is the measurable change in behaviour or outcome the work produces. The compression strips design thinking to three accountable artefacts, making it usable in social-sector and policy contexts where 5-stage process diagrams felt foreign (IDEO.org, 2015).

WHO DEVELOPED IT

IDEO.org, the non-profit arm of IDEO, codified the 3-i framing in *The Field Guide to Human-Centered Design* (2015). Synthesises IDEO's 1990s human-centered-design tradition (Kelley, 2001) with the social-sector evaluation literature on theory-of-change models.

WHY IT WAS DEVELOPED

Social-sector partners — NGOs, foundations, governments — often found the 5-stage process diagram (Plattner, 2010) too rigid for political-and-funding contexts. IDEO.org compressed the work to three accountable artefacts that mapped cleanly onto theory-of-change frameworks funders already used, making design thinking adoptable in social impact.

HOW IT IS USED

Document each project against three artefacts: the Insight (a single specific observation), the Idea (the design response), and the Impact (the measured change). Each must be tied to evidence; abstract claims at any stage signal that more research is needed.

WHEN IT IS USED

Use in social-impact innovation, in NGO and foundation programme design, when working with funders who require theory-of-change documentation, and as a pitch-and-reporting compression of longer DT engagements.

BY WHOM IT IS USED

Social entrepreneurs, NGO programme leads, public-sector innovation teams, foundation programme officers, and impact-investing analysts.

FIGURE 1 · THREE STEPS

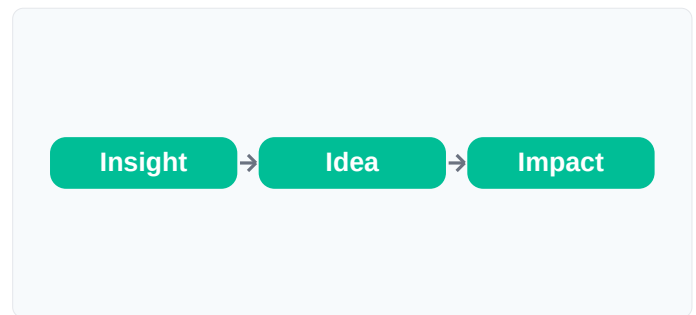


Figure 1. Three accountable artefacts; each must be tied to evidence. Adapted from IDEO.org (2015).

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Where the Stanford 5-stage (2010) provided detailed teaching modes and Brown's (2008) Triple Loop framed overlapping spaces, the 3-i Framework's contribution is the maximum-compression form for accountability contexts. Building on Weiss's (1995) theory-of-change tradition, it bridges design thinking and impact evaluation.

RELATED FRAMEWORKS IN THIS COURSE

DVF Framework · Slibrary 1

Brown's Triple Loop · Slibrary 1

Stanford 5-Stage · Slibrary 1

Double Diamond · Slibrary 1

SOURCES (HARVARD REFERENCING STYLE)

IDEO.org (2015) *The Field Guide to Human-Centered Design*. San Francisco: IDEO.org.

Kelley, T. (2001) *The Art of Innovation*. New York: Doubleday.

Weiss, C.H. (1995) 'Nothing as Practical as Good Theory: Exploring Theory-Based Evaluation', in *New Approaches to Evaluating Community Initiatives*. Washington: Aspen Institute.

Brown, T. and Wyatt, J. (2010) 'Design Thinking for Social Innovation', *Stanford Social Innovation Review*, 8(1), pp. 30–35.

Liedtka's 4 Truths

SLIBRARY 2 · SHEET 1 / 5

Four cognitive constraints DT imposes on managerial decision-making

PROFESSIONAL DEFINITION

Jeanne Liedtka's empirical synthesis of why design thinking works argues that it imposes four productive constraints on managerial cognition: *immersion* in users' worlds (countering abstraction), *diversification* of ideas (countering premature convergence), *evidence-based experimentation* (countering opinion-led decision-making), and *collaboration* across silos (countering departmental optimisation). Each "truth" addresses a documented decision-making bias; together they constitute the discipline, not the toolkit (Liedtka, 2018).

WHO DEVELOPED IT

Jeanne M. Liedtka, professor at the Darden School of the University of Virginia, in her 2018 *Harvard Business Review* article "Why Design Thinking Works". Built on six years of multi-organisation case research at hospitals, public-sector agencies and corporates documented in *Design Thinking for the Greater Good* (Liedtka, Salzman and Azer, 2017).

WHY IT WAS DEVELOPED

Practitioners taught design thinking as tools and processes; sceptics dismissed it as ritual. Liedtka argued that the consistent organisational benefits weren't from any specific tool but from four cognitive shifts the practice forced — making the case for design thinking on managerial-decision-quality grounds rather than aesthetic ones.

HOW IT IS USED

Audit each major decision against the four truths: did we immerse in users' realities? Did we diversify ideas before converging? Did we test with evidence? Did we collaborate across silos? Each "no" signals where the decision is most likely to fail.

WHEN IT IS USED

Use in senior-leadership coaching, in transformation-programme design, in governance reviews of strategic decisions, and as the management-grade case for design-thinking investment.

BY WHOM IT IS USED

Senior managers, organisational-development leaders, strategy consultants, and academic researchers studying design-thinking outcomes empirically.

FIGURE 1 · FOUR COGNITIVE SHIFTS

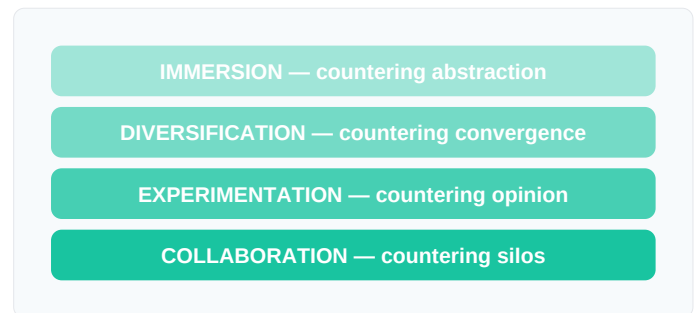


Figure 1. Each truth counters a documented decision-making bias. Adapted from Liedtka (2018).

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Where Brown (2008) defined design thinking as a mindset and Plattner (2010) operationalised it as a process, Liedtka's contribution is the empirical-mechanism account: *here are the four cognitive shifts that produce the organisational outcomes*. Building on Kahneman's (2011) decision-bias research, it grounds DT in cognitive psychology.

RELATED FRAMEWORKS IN THIS COURSE

3-i Framework · Slibrary 1

Dosi's 10-Dim Mindset · Slibrary 1

Mindset · Process · Toolbox · Slibrary 1

Kelley's Creative Confidence · Slibrary 1

SOURCES (HARVARD REFERENCING STYLE)

- Liedtka, J. (2018) 'Why Design Thinking Works', *Harvard Business Review*, 96(5), pp. 72–79.
 Liedtka, J., Salzman, R. and Azer, D. (2017) *Design Thinking for the Greater Good*. New York: Columbia Business School Publishing.
 Liedtka, J. (2015) 'Perspective: Linking Design Thinking with Innovation Outcomes', *Journal of Product Innovation Management*, 32(6), pp. 925–938.
 Kahneman, D. (2011) *Thinking, Fast and Slow*. New York: Farrar, Straus and Giroux.

Dosi's 10-Dim Mindset

SLIBRARY 2 · SHEET 2 / 5

Empirically validated 10-dimension instrument for measuring DT mindset

PROFESSIONAL DEFINITION

Clio Dosi and colleagues' Design Thinking Mindset (DTM) is the first empirically validated psychometric instrument operationalising what "thinking like a designer" actually means. Ten dimensions — including *tolerance for ambiguity*, *empathy*, *openness to diverse perspectives*, *learning orientation*, *systemic vision*, *experimentation orientation*, *embracing risk*, *mindfulness*, *critical questioning* and *self-awareness* — are scored via a validated 71-item questionnaire. Scores predict design-thinking outcomes and identify development gaps (Dosi, Rosati and Vignoli, 2018).

WHO DEVELOPED IT

Clio Dosi, Federica Rosati and Matteo Vignoli at the University of Bologna, in their 2018 paper for the *DESIGN 2018* conference. Built on extensive prior literature on creativity (Amabile, 1996), tolerance for ambiguity (Furnham and Ribchester, 1995), and design-thinking dispositions.

WHY IT WAS DEVELOPED

Practitioners and researchers debated what design-thinking mindset was without measurement; the debate was unresolvable. Dosi et al. produced the first instrument that turned mindset into a measurable variable researchers could correlate with outcomes and trainers could use to identify development needs.

HOW IT IS USED

Administer the 71-item DTM questionnaire (5-point Likert scale). Score by dimension. Aggregate to overall DTM score. Compare against domain norms; identify low-scoring dimensions as targeted development priorities.

WHEN IT IS USED

Use in design-thinking training programme assessment, in pre/post-workshop measurement, in academic research correlating mindset with outcomes, and in talent-selection contexts where DT mindset is a hiring criterion.

BY WHOM IT IS USED

Design-thinking educators, organisational-development consultants, HR analytics teams, academic researchers, and corporate-innovation programme directors measuring talent development.

FIGURE 1 · TEN DIMENSIONS

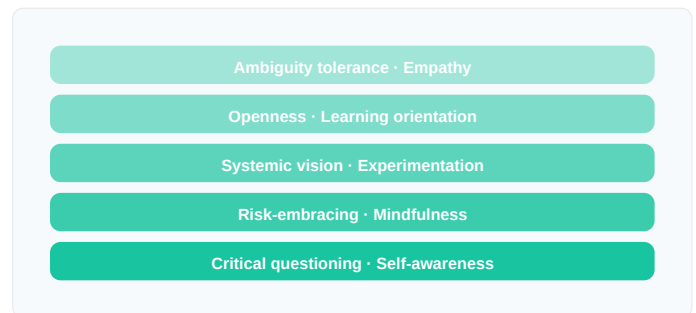


Figure 1. Ten dimensions, 71 items; scores correlate with design-thinking outcomes. Adapted from Dosi et al. (2018).

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Where Liedtka (2018) qualitatively characterised four DT cognitive shifts and earlier creativity research (Amabile, 1996) measured general creative disposition, Dosi et al.'s contribution is the DT-specific psychometric instrument. Building on Furnham and Ribchester's (1995) ambiguity-tolerance tradition, it makes DT mindset rigorously measurable.

RELATED FRAMEWORKS IN THIS COURSE

Liedtka's 4 Truths · Slibrary 1

Kelley's Creative Confidence · Slibrary 1

Mindset · Process · Toolbox · Slibrary 1

3-i Framework · Slibrary 1

SOURCES (HARVARD REFERENCING STYLE)

Dosi, C., Rosati, F. and Vignoli, M. (2018) 'Measuring Design Thinking Mindset', in *Proceedings of the DESIGN 2018 15th International Design Conference*. Dubrovnik: The Design Society, pp. 1991–2002.

Amabile, T.M. (1996) *Creativity in Context*. Boulder: Westview Press.

Furnham, A. and Ribchester, T. (1995) 'Tolerance of Ambiguity: A Review of the Concept, Its Measurement and Applications', *Current Psychology*, 14(3), pp. 179–199.

Liedtka, J. (2018) 'Why Design Thinking Works', *Harvard Business Review*, 96(5), pp. 72–79.

Kelley's Creative Confidence

SLIBRARY 2 · SHEET 3 / 5

Tom & David Kelley — *creativity is a learnable skill, not a birthright*

PROFESSIONAL DEFINITION

Tom and David Kelley's Creative Confidence is the position that creativity is not a fixed trait granted to some and denied to others, but a skill anyone can develop through structured practice — the way confidence in any domain is built. The Kelleys' research at IDEO and Stanford documents how guided exposure to small creative wins compounds into genuine creative capability over months. The frame is foundational: without it, design-thinking training fails because students self-categorise as "not creative" (Kelley and Kelley, 2013).

WHO DEVELOPED IT

Tom Kelley (IDEO) and David Kelley (Stanford d.school founder) in their 2013 book *Creative Confidence*, drawing on three decades of IDEO consulting practice and Stanford d.school teaching. Built on Bandura's (1977) self-efficacy theory and Csikszentmihalyi's (1996) creativity research.

WHY IT WAS DEVELOPED

Most adults — including senior managers — self-classified as "not creative" and stopped engaging at the first creative friction. The Kelleys argued — backed by neuroscience and self-efficacy research — that this self-classification was empirically wrong, and that small structured wins could systematically rebuild creative confidence in adults.

HOW IT IS USED

Use guided exposure: assign small, low-stakes creative tasks (sketching ideas, asking unfamiliar questions, prototyping crudely). Acknowledge each completion publicly. Compound success over weeks. Measure changes in self-reported creative confidence pre and post.

WHEN IT IS USED

Use in design-thinking onboarding, in adult-learning contexts, in creative-team rebuilding after failure, and as the foundational mindset module before any design-thinking tool training.

BY WHOM IT IS USED

Design educators, organisational-development consultants, executive coaches, MBA capstone instructors, and HR business partners running creative-skills training.

FIGURE 1 · CONFIDENCE COMPOUNDS

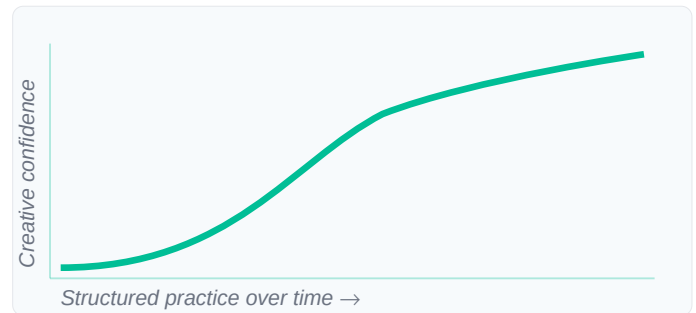


Figure 1. Small structured wins compound; the gap between "creative" and "non-creative" people closes with practice.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Where Csikszentmihalyi (1996) studied creativity at the genius level and Amabile (1996) measured creative output abstractly, the Kelleys' contribution is the prescriptive adult-learning protocol for rebuilding creative confidence. Building on Bandura's (1977) self-efficacy tradition, it gives DT trainers a foundational mindset intervention.

RELATED FRAMEWORKS IN THIS COURSE

Liedtka's 4 Truths · Slibrary 1

Dosi's 10-Dim Mindset · Slibrary 1

Mindset · Process · Toolbox · Slibrary 1

Brown's Triple Loop · Slibrary 1

SOURCES (HARVARD REFERENCING STYLE)

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- Kelley, T. and Kelley, D. (2012) 'Reclaim Your Creative Confidence', *Harvard Business Review*, 90(12), pp. 115–118.
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- Csikszentmihalyi, M. (1996) *Creativity: Flow and the Psychology of Discovery and Invention*. New York: HarperCollins.

Buchanan's Wicked Problems

SLIBRARY 2 · SHEET 4 / 5

Design's domain is the wicked problems that have no stopping rule

PROFESSIONAL DEFINITION

Richard Buchanan's foundational 1992 essay locates design's distinctive contribution in *wicked problems* — problems with no definitive formulation, no stopping rule, no true-or-false solutions, and where every solution-attempt changes the problem itself. Wicked problems resist the linear problem-solve approach of engineering and management. Design's iterative, prototype-driven, frame-shifting practice is the only mode that makes traction on them — making design not a styling discipline but a method for the genuinely intractable (Buchanan, 1992; Rittel and Webber, 1973).

WHO DEVELOPED IT

Richard Buchanan, design scholar at Carnegie Mellon, in his 1992 *Design Issues* paper "Wicked Problems in Design Thinking". Built directly on Horst Rittel and Melvin Webber's (1973) original "wicked problems" framing in policy planning.

WHY IT WAS DEVELOPED

Design was widely treated as cosmetic — applied after "real" problem-solving was complete. Buchanan repositioned design as the only discipline equipped for the problems that systematically defeat linear approaches: ill-defined, value-laden, interconnected with other problems, mutating under intervention.

HOW IT IS USED

Diagnose problems against Rittel and Webber's ten characteristics: definitive formulation, stopping rule, true/false solutions, immediate-and-ultimate test, one-shot operation, exhaustive solution-set, uniqueness, symptom-of-other-problem, explanation-discrepancy, no-right-to-be-wrong. Wicked problems require designerly methods.

WHEN IT IS USED

Use in policy-design, social-innovation, organisational-transformation, climate-and-sustainability contexts, and in any situation where stakeholders disagree fundamentally about what the problem even is.

BY WHOM IT IS USED

Public-policy designers, social entrepreneurs, organisational-development consultants, transformation officers, academic researchers in design and policy.

FIGURE 1 · TEN WICKED PROPERTIES

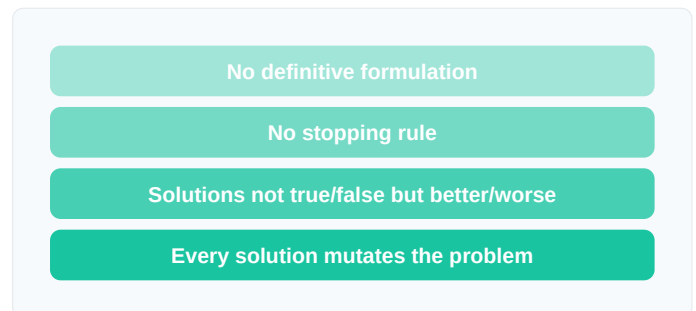


Figure 1. Wicked problems resist linear problem-solving; design's iterative practice is the only mode that gains traction.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Where Rittel and Webber (1973) named wicked problems in policy contexts and Simon (1969) framed design as iterative problem-solving generally, Buchanan's contribution is connecting design specifically to the wicked problem class. Building on the broader systems-thinking tradition, he made the case for design as a primary management discipline, not a styling adjunct.

RELATED FRAMEWORKS IN THIS COURSE

Brown's Triple Loop · Slibrary 1

Mindset · Process · Toolbox · Slibrary 1

Liedtka's 4 Truths · Slibrary 1

Dosi's 10-Dim Mindset · Slibrary 1

SOURCES (HARVARD REFERENCING STYLE)

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Mindset · Process · Toolbox

SLIBRARY 2 · SHEET 5 / 5

Brenner & Tschepe — DT is mindset and process and toolbox simultaneously

PROFESSIONAL DEFINITION

Brenner and Tschepe's tripartite model argues that design thinking cannot be reduced to any single layer — it is a *mindset* (disposition towards ambiguity, user-centricity, iteration), a *process* (the diverge-converge iterative loop), and a *toolbox* (specific techniques: empathy maps, prototyping, journey-mapping), always simultaneously. Teaching one layer without the others produces ritual, not capability — students who execute the toolbox without the mindset, or hold the mindset without the techniques, miss most of the value (Brenner, Uebernicket and Abrell, 2016; Tschepe, 2017).

WHO DEVELOPED IT

Walter Brenner, Falk Uebernicket and Thomas Abrell at the University of St Gallen, in their 2016 chapter for the *Design Thinking for Innovation* Springer volume. Reinforced and extended by Samuel Tschepe's 2017 paper "Mindset, Process, Method: A Comprehensive Descriptive Model for Design Thinking".

WHY IT WAS DEVELOPED

Design-thinking education was fragmenting into camps — mindset-only retreats that produced no concrete output, toolbox-only workshops that produced concrete but unfit output, process-only stage-gate diagrams that lost the iterative spirit. Brenner and Tschepe argued the three layers were inseparable in practice, and teaching them apart was teaching them wrong.

HOW IT IS USED

Audit each design-thinking intervention against the three layers: is the mindset present (real user contact, comfort with ambiguity, iterative posture)? Is the process structure present (diverge-converge, evidence-based gates)? Is the toolbox being used appropriately (right tool, right phase)? Missing any layer signals incomplete capability.

WHEN IT IS USED

Use in design-thinking programme design, in coaching-supervision contexts, in capability assessment of innovation-programme directors, corporate-innovation teams, and as a diagnostic for why a DT intervention failed despite using the right tools.

BY WHOM IT IS USED

Design-thinking educators, corporate learning-and-development leaders, innovation-programme directors, academic researchers in design education.

FIGURE 1 · THREE INSEPARABLE LAYERS

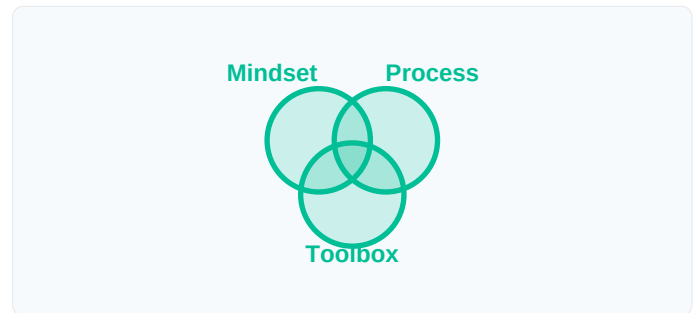


Figure 1. Teaching one layer without the others produces ritual. Adapted from Brenner et al. (2016).

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Where Brown (2008) emphasised the mindset and Plattner (2010) emphasised the process, Brenner and Tschepe's contribution is the explicit insistence on simultaneity — none of the three layers reduces to the others. Building on educational-theory traditions of integrated knowledge-skills-attitudes, it gives design-thinking education a coherent diagnostic frame.

RELATED FRAMEWORKS IN THIS COURSE

Brown's Triple Loop · Slibrary 1

Liedtka's 4 Truths · Slibrary 1

Dosi's 10-Dim Mindset · Slibrary 1

Buchanan's Wicked Problems · Slibrary 1

SOURCES (HARVARD REFERENCING STYLE)

Brenner, W., Uebernicket, F. and Abrell, T. (2016) 'Design Thinking as Mindset, Process, and Toolbox', in *Design Thinking for Innovation*. Cham: Springer, pp. 3–21.

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Plattner, H., Meinel, C. and Leifer, L. (eds.) (2011) *Design Thinking: Understand – Improve – Apply*. Berlin: Springer.

Brown, T. (2009) *Change by Design*. New York: HarperBusiness.

Empathy Map

SLIBRARY 3 · SHEET 1 / 5

Says · Thinks · Does · Feels — externalise the user's lived experience

PROFESSIONAL DEFINITION

The Empathy Map is a four-quadrant canvas that externalises a user's experience along four dimensions: *Says* (verbatim quotes), *Thinks* (inferred internal state), *Does* (observed behaviour), *Feels* (emotional registers). Populating all four from real research forces the team past surface demographic-and-feature thinking into the texture of the user's lived experience — the foundation of empathy-driven design. Originally developed at

XPLANE, the empathy map has become a near-universal DT artefact (Gray, Brown and Macanufo, 2010; Bland, 2017).

Dave Gray and the XPLANE team in the 2000s, codified in *Gamestorming* (Gray, Brown and Macanufo, 2010); refined into the modern five-quadrant version (adding Pains and Gains) by David Bland (2017) and the Strategyzer team. Roots in ethnographic-fieldwork tradition (Geertz, 1973).

WHY IT WAS DEVELOPED

Personas (Cooper, 1999) were too abstract to drive design; verbatim transcripts were too long. The Empathy Map sits between — concrete enough to hold specific user evidence, structured enough to compare across users, fast enough to populate in a workshop hour.

HOW IT IS USED

Pick a target user. Populate the four quadrants from research evidence, not assumption. Each quadrant entry must trace to a specific interview, observation or artefact. Identify contradictions between quadrants (e.g., what they Say vs what they Do) — these are the most valuable design insights.

WHEN IT IS USED

Use after discovery-interview research, in empathy-phase workshops, before persona consolidation, and as the lens for testing whether the team has genuine user understanding versus assumed understanding.

BY WHOM IT IS USED

UX researchers, design-thinking facilitators, product managers, MBA design-thinking courses, and B2B account managers studying buyer-stakeholder experience.

FIGURE 1 · FOUR QUADRANTS



Figure 1. Contradictions across quadrants are the most valuable design insights.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Where Cooper's (1999) personas synthesised abstract user archetypes and ethnographic transcripts captured raw verbatim, the Empathy Map's contribution is the structured four-quadrant compression. Building on Geertz's (1973) thick-description tradition, it makes empathy a workshop-actionable artefact.

RELATED FRAMEWORKS IN THIS COURSE

Personas · Slibrary 1

Customer Journey Map · Slibrary 1

5 Whys · Slibrary 1

Stanford 5-Stage · Slibrary 1

SOURCES (HARVARD REFERENCING STYLE)

Gray, D., Brown, S. and Macanufo, J. (2010) *Gamestorming: A Playbook for Innovators, Rulebreakers, and Changemakers*. Sebastopol: O'Reilly.

Bland, D. (2017) 'The Empathy Map (2017 Update)', *Strategyzer blog*.

Cooper, A. (1999) *The Inmates Are Running the Asylum*. Indianapolis: Sams.

Geertz, C. (1973) *The Interpretation of Cultures*. New York: Basic Books.

Personas

SLIBRARY 3 · SHEET 2 / 5

Composite user archetypes grounded in interview evidence

PROFESSIONAL DEFINITION

Alan Cooper's Personas technique translates research into composite user archetypes that carry forward into design decisions. Effective personas are grounded in interview evidence, given a specific goal and a specific frustration, and used as a decision-making lens — not as marketing demographics. Bad personas ("Sarah, 34, suburban mom, drives a minivan") are demographic shells that no design team actually consults; good personas ("Maya needs to onboard 3 contractors before Friday but doesn't trust the new tool") drive specific feature decisions (Cooper, 1999, 2004).

WHO DEVELOPED IT

Alan Cooper, software interaction design pioneer, in *The Inmates Are Running the Asylum* (1999). The persona technique was elaborated in *About Face* (Cooper, 2004) and has been refined through dozens of practitioner texts. Roots in market-research segmentation and ethnographic fieldwork.

WHY IT WAS DEVELOPED

Designers asked to design for "users" defaulted to designing for themselves; product managers asked to specify "the customer" produced ambiguous specifications. Cooper's personas constrained the design conversation to specific archetypal users with goals and frustrations, blocking self-projection and ambiguous specification.

HOW IT IS USED

Build each persona from 5+ interviews of similar users. Give them a specific name, goal, key frustration, current workaround, and one-line rationale. Use only personas as the audience for design decisions; "Maya wouldn't use this" is decisive in a way that "users wouldn't use this" never is.

WHEN IT IS USED

Use after empathy research consolidates patterns, in product-management roadmap reviews, in feature-prioritisation discussions, and as the lens for testing whether design decisions are user-grounded.

BY WHOM IT IS USED

UX designers, product managers, design-thinking teams, B2B account managers, and marketing strategists. Standard in software, financial services and healthcare design contexts.

FIGURE 1 · PERSONA ANATOMY

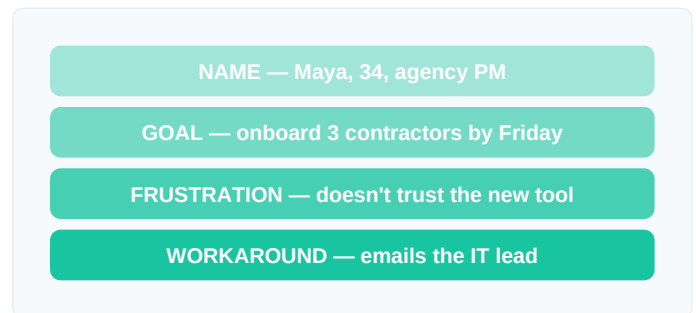


Figure 1. Each persona is concrete enough to drive specific feature decisions. Adapted from Cooper (1999).

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Where market-research segmentation grouped users statistically and ethnographic transcripts captured individual users, Cooper's contribution is the composite-archetype synthesis usable as a design-decision lens. Building on Maslow's (1954) needs-hierarchy tradition, it gave design teams concrete user-fictions for decision-making.

RELATED FRAMEWORKS IN THIS COURSE

[Empathy Map · Slibrary 1](#)
[Customer Journey Map · Slibrary 1](#)
[5 Whys · Slibrary 1](#)
[How Might We \(HMW\) · Slibrary 1](#)

SOURCES (HARVARD REFERENCING STYLE)

Cooper, A. (1999) *The Inmates Are Running the Asylum: Why High-Tech Products Drive Us Crazy and How to Restore the Sanity*. Indianapolis: Sams.
 Cooper, A., Reimann, R. and Cronin, D. (2004) *About Face 3: The Essentials of Interaction Design*. Indianapolis: Wiley.
 Pruitt, J. and Adlin, T. (2006) *The Persona Lifecycle*. San Francisco: Morgan Kaufmann.
 Goodwin, K. (2009) *Designing for the Digital Age*. Indianapolis: Wiley.

Customer Journey Map

SLIBRARY 3 · SHEET 3 / 5

Stages × touchpoints × emotions across the user's full path

PROFESSIONAL DEFINITION

The Customer Journey Map is a structured visualisation of a customer's complete experience across stages (Awareness → Consideration → Purchase → Use → Advocacy), touchpoints (each interaction with the product or service) and emotional registers (frustration, anxiety, delight). The map externalises a journey usually invisible to the firm because no single department sees the whole; surfacing it makes pain points and white-space opportunities visible. Originated in service-design practice; now standard in product, marketing and CX (Stickdorn et al., 2018; Shostack, 1984).

WHO DEVELOPED IT

G. Lynn Shostack pioneered service-blueprint diagrams in 1984. The modern customer-journey-map form was operationalised in service-design practice through the 2000s and codified in Stickdorn et al.'s *This Is Service Design Doing* (2018). Heritage in Womack and Jones's (1996) value-stream-mapping tradition.

WHY IT WAS DEVELOPED

Departments optimised individual touchpoints (marketing → ads, sales → demos, support → tickets) without seeing the cumulative customer experience. Journey maps revealed that customers experienced the cumulative experience, and that local-departmental optimisation often degraded the whole. The map made the whole visible.

HOW IT IS USED

Map stages along the horizontal axis. Plot touchpoints in each stage. Plot the customer's emotional state in each stage (curve from positive to negative). Identify pain points (deepest negative valleys) and moments of truth (highest-stakes touchpoints). Design interventions target these specific points.

WHEN IT IS USED

Use in service-design redesign, in customer-experience strategy reviews, in marketing-funnel optimisation, in healthcare-and-financial-services patient/customer experience design, and as the diagnostic before any large CX investment.

BY WHOM IT IS USED

Service designers, customer-experience leaders, product managers, marketing strategists, and operations leaders running CX transformation programmes.

FIGURE 1 · EMOTIONAL TRAJECTORY

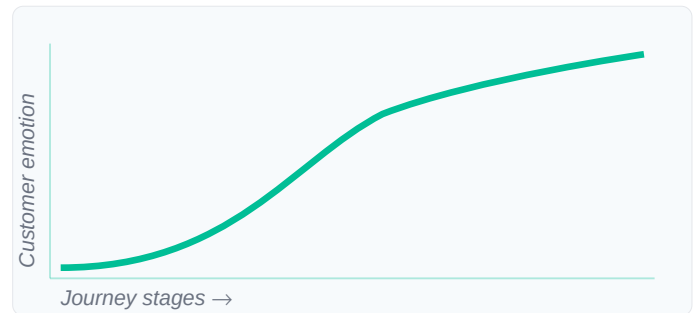


Figure 1. Pain points are valleys; moments of truth are highest-stakes peaks. Both demand intervention.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Where Shostack's (1984) service blueprint focused on operational steps and Womack and Jones's (1996) value-stream mapped manufacturing flow, the Customer Journey Map's contribution is the customer's *emotional* trajectory across stages and touchpoints. Building on Norman's (2004) emotional-design tradition, it gives CX a unified diagnostic.

RELATED FRAMEWORKS IN THIS COURSE

Empathy Map · Slibrary 1

Personas · Slibrary 1

How Might We (HMW) · Slibrary 1

Job Map · Slibrary 5

SOURCES (HARVARD REFERENCING STYLE)

Stickdorn, M., Hormess, M.E., Lawrence, A. and Schneider, J. (2018) *This Is Service Design Doing*. Sebastopol: O'Reilly.
 Shostack, G.L. (1984) 'Designing Services That Deliver', *Harvard Business Review*, 62(1), pp. 133–139.
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5 Whys

SLIBRARY 3 · SHEET 4 / 5

Toyota's recursive root-cause technique — five iterations gets you to structure

PROFESSIONAL DEFINITION

The 5 Whys is Toyota's recursive root-cause analysis technique: when a problem occurs, ask "why?", answer it, then ask "why?" of that answer, and so on five times. The technique systematically descends from symptoms through proximate causes to structural root causes. The exact number five is empirical, not magic — it usually marks the depth at which fixable structural causes become visible, beneath which speculation begins. Developed for manufacturing problem-solving, now standard in DT and lean (Ohno, 1988; Liker, 2004).

WHO DEVELOPED IT

Sakichi Toyoda, founder of Toyota Industries, in the 1930s; codified by Taiichi Ohno in *Toyota Production System* (1988) as one of the core tools of the Toyota method. Adopted across lean manufacturing, agile development and design-thinking traditions through the 1990s–2000s.

WHY IT WAS DEVELOPED

Manufacturing-floor problem-solving frequently stopped at proximate causes, producing fixes that addressed symptoms while structural problems persisted and recurred. Toyoda's discipline forced root-cause depth: by the fifth "why", teams had typically descended past the symptomatic layer into the structural one.

HOW IT IS USED

State the problem precisely. Ask "why does this happen?" — write the answer. Ask "why?" of that answer — write the next. Repeat to depth five (or until you stop being able to honestly answer). The deepest answer is the structural cause; intervention there prevents recurrence.

WHEN IT IS USED

Use in incident post-mortems, in design-thinking define-phase work, in lean problem-solving, in software-engineering retrospectives, and whenever a recurring problem is suspected to have structural causes.

BY WHOM IT IS USED

Operations managers, manufacturing engineers, agile coaches, design-thinking facilitators, and quality leaders running root-cause analysis processes.

FIGURE 1 · FIVE ITERATIONS DOWN

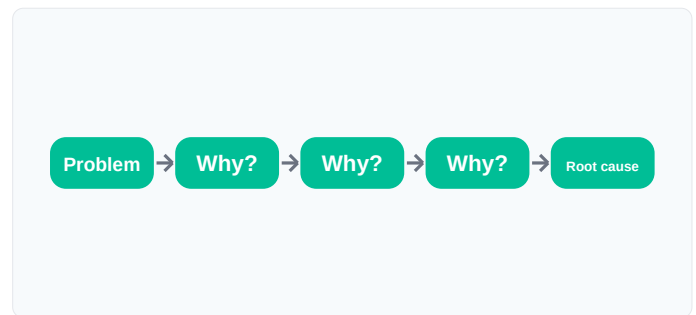


Figure 1. Five iterations typically descend past symptoms to structural causes. Adapted from Ohno (1988).

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Where Western root-cause analysis (Ishikawa fishbone, FMEA) used breadth-first decomposition, Toyoda's contribution is depth-first recursive interrogation. Building on Eastern philosophical traditions of recursive questioning, the 5 Whys made root-cause depth a workshop-fast discipline.

RELATED FRAMEWORKS IN THIS COURSE

Empathy Map · Slibrary 1

Customer Journey Map · Slibrary 1

How Might We (HMW) · Slibrary 1

Pre-Mortem · Slibrary 4

SOURCES (HARVARD REFERENCING STYLE)

Ohno, T. (1988) *Toyota Production System: Beyond Large-Scale Production*. Cambridge, MA: Productivity Press.

Liker, J.K. (2004) *The Toyota Way: 14 Management Principles from the World's Greatest Manufacturer*. New York: McGraw-Hill.

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Womack, J.P., Jones, D.T. and Roos, D. (1990) *The Machine That Changed the World*. New York: Rawson Associates.

How Might We (HMW)

SLIBRARY 3 · SHEET 5 / 5

The generative question that re-opens a problem just enough to invite ideation

PROFESSIONAL DEFINITION

How Might We is the generative-question format that re-opens a problem just enough to invite ideation. The phrasing is precise: *How* presumes solvability; *Might* invites multiple options without committing; *We* implies collective ownership. HMW questions are calibrated — neither too narrow ("how might we add a button?") nor too broad ("how might we improve education?") — at the abstraction level where divergent ideation is most productive. The form was introduced by Min Basadur at Procter & Gamble and popularised through IDEO (Berger, 2012; Liedtka and Ogilvie, 2011).

WHO DEVELOPED IT

Min Basadur at Procter & Gamble in the 1970s, codified in his Simplex creative-problem-solving system; popularised in design thinking through Tim Brown's writing (Brown, 2009) and IDEO's facilitation playbook. Documented for the broader audience in Warren Berger's *A More Beautiful Question* (2012).

WHY IT WAS DEVELOPED

Workshop facilitators found that the abstraction level of the framing question determined ideation productivity. Too-narrow questions produced incremental fixes; too-broad questions produced abstraction. Basadur's HMW form sat at the productive middle — calibrated to invite multiple specific directions without committing to any.

HOW IT IS USED

Take the problem statement. Reframe as "How might we...?". Test the abstraction level: can you generate at least 10 distinct directions? If fewer, broaden. Can you generate concrete prototypable options? If not, narrow. Iterate the framing until ideation flows.

WHEN IT IS USED

Use in the transition from Define to Ideate phase, before any brainstorming or Crazy 8s session, when teams have stalled on ideation, and as the central reframing move of design-thinking workshops.

BY WHOM IT IS USED

Design-thinking facilitators, workshop coaches, product managers, innovation programme leads, and MBA capstone instructors.

FIGURE 1 · THREE-WORD CALIBRATION



Figure 1. Each word does specific structural work; the calibrated abstraction is the design.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Where Osborn's (1953) original brainstorming gave rules for generating ideas and Polya's (1945) heuristics for problem-solving asked questions abstractly, Basadur's contribution is the calibrated framing form for design ideation specifically. Building on Osborn-Parnes Creative Problem Solving (1967), HMW became the canonical DT reframing tool.

RELATED FRAMEWORKS IN THIS COURSE

[5 Whys · Slibrary 1](#)
[Customer Journey Map · Slibrary 1](#)
[Empathy Map · Slibrary 1](#)
[Stanford 5-Stage · Slibrary 1](#)

SOURCES (HARVARD REFERENCING STYLE)

- Berger, W. (2012) 'The Secret Phrase Top Innovators Use', *Harvard Business Review* (online).
 Liedtka, J. and Ogilvie, T. (2011) *Designing for Growth*. New York: Columbia Business School Publishing.
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